

Stella M. Sigala

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PROFILE

MSc researcher in Applied Meteorology and Atmospheric Physics with a BSc in Theoretical Computational Physics and Astrophysics. MSc thesis: "Chasing after an observational butterfly effect in end-to-end AI-DOP models " (ongoing).

- Research focused on machine learning and data-driven forecasting systems
- Reanalysis and observation based model assessments
- Forecast verification and statistical evaluations
- Python data analysis of large atmospheric datasets
- Spatiotemporal analysis reports and error metrics

RESEARCH & PROJECTS

- *Analysis of atmospheric stability over the Island of Sardinia, Italy (February and July 2025).*
- *Wind magnitude and direction: A Jet-stream analysis using ERA5-Land and CERRA.*
- *Linking Jet Waviness, Blocking Position and Cold Spells: A January 2017 case study.*

TRAINING

ECMWF, Reading, UK, 2026: Completed training in numerical techniques used in weather prediction models for solving the governing equations of atmospheric dynamics on HPC systems. Examined the links and contrasts between established numerical weather prediction models and emerging machine learning approaches for weather forecasting.

WORK EXPERIENCE

Laboratory of Atmospheric Physics, University of Patras, Greece, 2025-2026: Supported undergraduate laboratory exercises in Meteorology and Atmospheric Physics.

Block, Patras, Greece, 2023: Climbing Instructor. Technique, safety, training and teaching ways of staying focused and calm in a competitive environment.

LANGUAGES

English (Native) • Greek (Native) • German (Beginner)